4th International Conference on Electrical Information and Communication Technology (EICT), 20-22 December 2019, Khulna, Bangladesh

# Stress Identification during Sustained Mental Task and Brain Relaxation Modeling with $\beta/\alpha$ Band Power Ratio

Md. Salah Uddin Yusuf, Monira Islam and Shidhartho Roy
*Department of Electrical and Electronic Engineering*
*Khulna Univeristy of Engineering & Technology (KUET)*
Khulna-9203, Bangladesh
ymdsalahu2@gmail.com, monira_kuet08@yahoo.com and swapno15roy@gmail.com

***Abstract*—Stress is the non-specific response of human brain which changes according to the demand of day to day life. In this paper stress and relaxation of brain is identified from the activation of alpha (α) and beta (β) EEG bands depending on their power spectral density (PSD). Stress occurs due to sustained mental task in a test (Bengali, English, Math) environment and the work has been performed in to Reading, Writing and Mathematical problem-solving stages. When $\beta/\alpha$ power ratio is 1.5 or more it indicates stress. The relaxation of the brain is estimated taking the PSD at different time interval after the brain activity and when the $\beta/\alpha$ ratio returns to the value of relax state then it is determined that relaxation is obtained. Finally brain relaxation is modeled with mathematical formulation based on $\beta/\alpha$ power ratio.**

***Keywords—EEG bands, stress, mental task, band power ratio, relaxation modeling***

## I. Introduction

The higher level of cognitive engagement in any cognitive process indicates mental workload which has a direct impact on the quality of learning process for the learners [1]. An optimal level of mental workload facilitates an efficient learning process whereas cognitive overload negatively affects the task performance and causes a change in human behavior and directs towards stress [2]. Real-time measurement of mental stress can assist to identify the optimal level of mental workload and it enhances the subjective performance on the learning or test environment [3].

Stress can be estimated from perceptual, behavioral and physical responses to sustained mental activities. Evaluation of these responses to stress involves subjective estimations and perceptions at different test environment. Self-report questionnaires are one of the most commonly used methods to measure ones' level of stress [4]. Evaluating the mental stress using questionnaires is a subjective method [5] and stress can be quantified from human bio-signals [6]. The stress can be evaluated from physical response from the changes of heart rate, respiratory rate, skin conductance, blood volume or temperature. EEG signal are subdivided in to several frequency bands with spectral analysis which are alpha bands (8-13 Hz), beta bands (14-30 Hz), delta (0.5-4 Hz) and theta (4-8 Hz) bands. In the EEG, increased EEG alpha wave activity has been associated with an increased level of relaxation, otherwise it decreases during stress condition [7]. Delta and theta band are activated during deep sleep or unconscious and for consciousness or mental stress the alpha and beta band activity are more. So, in this paper, beta and alpha activation is analyzed using PSD and $\beta/\alpha$ band power ratio is used as relaxation factor.

Many researchers are working on stress indication. To evoke cognitive loading mental arithmetic, decision making, memory retention, high workload, and multiple tasks solutions are taken previously as experiment. Most common experiments are Color-Stroop Task [8], Montreal Imaging Stress Task and basic arithmetic questions under time limits [9]. So WKY et.al developed a bio-marker in which mental workload based on the frontal activities were measured by a mobile single channel EEG system [3]. Authors in [10] demonstrated that alpha and theta activities have close relationship with cognitive performance and mental effort. Raghayachan *et al.* showed that event-related Theta activity increases with memory load and it decreases steeply after completion of the task [11]. However, these findings were obtained from conventional multi-channel EEG with wet electrodes. Authors in [12] showed the activity of EEG bands during different emotional condition and mathematical modeling of these states. But there is found little work on brain relaxation analysis and modeling.

In this work, subject's mental stress is evaluated from EEG signals in a test environment during sustained mental activities. Cognitive workload is responsible for the stress which is identified from the non-stationary EEG signals. Mental workload changes human mental responses so relaxation analysis is also performed depending on their beta to alpha band power ratio. The more alpha band activity indicates relaxation of the brain whereas beta activity inverses this. A mathematical model is necessary to unify the brain relaxation analysis in different Bengali, English and Math test environments. So, in this paper a mathematical model is developed for brain relaxation which is the new outcome than the previous one. Fig.1 shows the flow chart of the proposed approach.

The rest of this paper is organized as follows. Section II briefly introduces methodology of this proposed work. Experimental results and discussion justifying the PSD analysis for stress indication and mathematical modeling of brain relaxation are reported in Section IV. Finally, Section V concludes the result.

## II. Methodology

### A. Categorization of Mental Tasks

This work focus on the identification of stress due to sustained mental task in the test environment. So Bengali, English and Mathematics tests are considered as the mental task stimuli to determine the stress of the learners. These

978-1-7281-6040-5/19/$31.00 ©2019 IEEE

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on July 20,2025 at 02:40:20 UTC from IEEE Xplore. Restrictions apply.

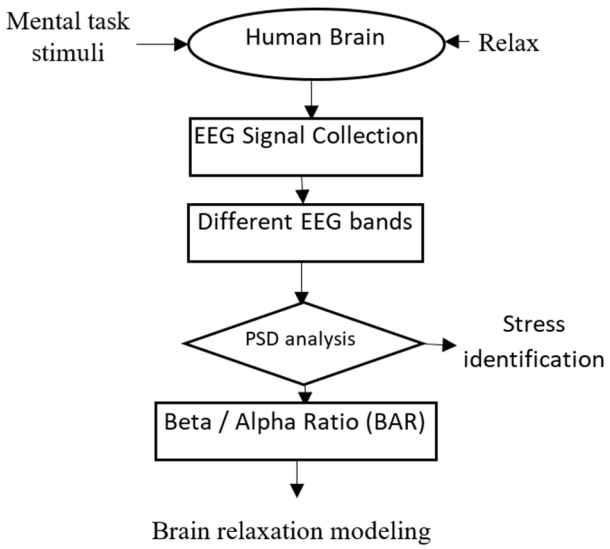



Fig. 1. Flow diagram of the proposed esearch approach.

TABLE I. CATEGORIZATION OF MENTAL TASK

| Test Environment | Task activities |
|---|---|
| Bengali | Reading |
| | Writing |
| Engilish | Reading |
| | Writing |
| Mathematics | Problem Reading |
| | Problem Solving |

three tests are subdivided in to reading, writing and problem-solving stages to identify the mental activities as shown in Table I.

### B. Subject Preparation

To select subjects, we have used random pick method. We have selected 52 subjects all aged from 20 to 23 (on average 22.3). All were university students. The very common thing for all the subjects is that Bengali is the native language whereas English is the second one. It is noteworthy that the medium of instruction is English for entire course curriculum in the universities. It is also notable that selected subjects do not suffer from sleep disorder or any other severe diseases. The subjects were observed around 7 weeks in the sealed laboratory room and around 18 hours of data are recorded on an average. Moreover, the subjects were not aware of the study objectives.

### C. EEG Signal Acquisition

EEG signals are collected in BME Lab, KUET using BIOPAC MP36 system. BIOPAC electrode lead set (SS2L), disposable vinyl electrodes (EL503), data collection unit (MP36 and MP150) with cable and power are required for acquisition the signal. The pictorial view of the EEG acquisition in the BME laboratory is shown in Fig. 2. The data acquisition protocol and the raw EEG signal with different EEG bands are shown in Figs. 3 and 4 respectively. One of the mentioning advantages of BIOPAC system is that it consists filtering to remove the power line carriers and divide the raw EEG signals in to different frequency bands (alpha, beta, delta and theta). For data collection process, at first in relaxed state 15 minute of data is recorded for further comparison. Then a mental task of 1-hour duration has been provided for observation. After 1 hour, immediately EEG data has been recorded. After 15 min. rest

Fig. 2. Pictorial view of EEG signal acquisition at BME lab KUET.

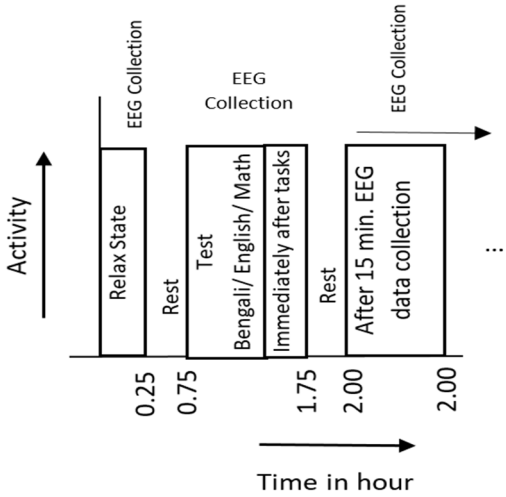



Fig. 3. Data acquisition protocol.

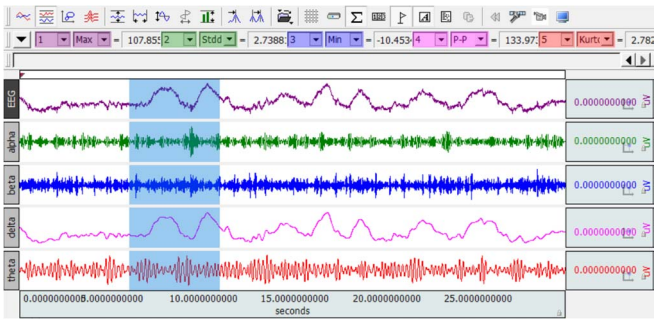



Fig. 4. Collected raw EEG signal with different frequency bands during sustained mental tasks.

again, EEG data has been recorded. In the same manner maintaining a regular interval EEG data has been recorded for brain relaxation analysis.

### D. PSD Analysis

Power spectral density (PSD) analysis is significant to identify the brain activation. The PSD value of any band shows the average power of the individual band in terms of frequency. The spectral analysis is performed to extract the PSD of the EEG bands. In this analysis, Hamming window is taken with zero padding and 1024 data samples are taken for extracting the value of PSD. Equation (1) and (2) represents the power spectral density function of alpha and beta band respectively where the average power is in the time range ($t_1$, $t_2$).

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on July 20,2025 at 02:40:20 UTC from IEEE Xplore. Restrictions apply.

$$\int_0^F PSD_{alpha}(k)dk = \int_0^F \frac{2|X(k_{alpha})|^2}{(t_2-t_1)} dk \quad (1)$$

$$\int_0^F PSD_{beta}(k)dk = \int_0^F \frac{2|X(k_{beta})|^2}{(t_2-t_1)} dk \quad (2)$$

Where, the symbol 'k' represents the variable time. The method FFT analysis gives us opportunity to analyze alpha and beta band. It is given in (3).

$$X_{alpha/beta}(k) = \{X(0), X(1), \ldots X(n-1)\} \quad (3)$$

*E. Brain Relaxation modeling*

Brain relaxation analysis is also performed with the value of PSD over time of alpha and beta bands after the completion of sustained mental tasks. The beta to alpha ratio (BAR) is obtained from the PSD analysis and taken as the brain relaxation factor. In this paper, BAR has been recorded over time and compared with the BAR of relax state. The graph is plotted with the BAR over different time interval and mathematical model is developed with least square algorithm in MATLAB platform which is reliable to fit the non- linear data. It is based on the calculation of the total square error as (4).

$$\sum_k (\mathrm{y_k} - f(x_k))^2 = \sum_k (y_k - a - bx_k - cx_k^2)^2 \quad (4)$$

which depends on three unknown parameters a, b and c. The values must be found out for which it becomes minimum. The proper assumptions from asserting a plot of data are show in (5).

$$\begin{cases} 0 = \frac{\partial}{\partial a} E(a,b,c)^2 = -2\sum_k (\mathrm{y_k} - a - bx_k - cx_k^2)^2 \\ 0 = \frac{\partial}{\partial b} E(a,b,c)^2 = -2\sum_k \mathrm{x_k}(\mathrm{y_k} - a - bx_k - cx_k^2)^2 \\ 0 = \frac{\partial}{\partial c} E(a,b,c)^2 = -2\sum_k x_k^2(\mathrm{y_k} - a - bx_k - cx_k^2)^2 \end{cases} \quad (5)$$

So, it becomes a linear equation in the unknowns a, b and c which provides the solution and provides a nearly quadratic equation.

## III. Results and Discussion

*A. Stress detection from PSD*

PSD analysis shows the activation of EEG bands due to different test environments. Table II shows the PSD based band activation. The delta and theta band are activated for deep sleep or unconsciousness stage of the brain. So, in this analysis alpha and beta activation is analyzed. Firstly, when the subjects were kept in relaxed state the alpha activation is 66% and $\beta/\alpha$ band power ratio is 0.509.

For Bengali, English and mathematical task the beta activity was greater than the alpha and the $\beta/\alpha$ ratio is approximately on average 1.5. The larger beta activity shows the stress condition of the brain. Fig. 5 shows the percentage of alpha and beta activity for relax stage and also for Bengali, English and Mathematics test environment.

*B. Brain relaxation Analysis with PSD*

Brain relaxation is analyzed with $\beta/\alpha$ band power ratio. Table III shows that the $\beta/\alpha$ band power ratio is decreasing over time which indicates that the beta activity is reducing whereas the alpha activity of brain is increasing so relaxation is obtained when the beta to alpha band power ratio returns to the value of the relax state. In this analysis $\beta/\alpha$ Ratio is taken as the relaxation factor. At the stage of relax the $\beta/\alpha$ Ratio is 0.509. The ratio is approximately 1.5 when the signal was taken is immediately after task. From the Table III, it is determined that after 2.5-hour brain is returning to the original relax state.

In the case of brain relaxation English seems more responsive to return to relax state than the other two test environment. The subjects were all the university students where the medium of instruction was English so the learners' brains are more comfortable with English than Bengali or Mathematics. Fig. 6 shows the brain relaxation curve over time which shows that Brain takes less time to be relaxed for English test and more time for Mathematics test.

*C. Mathematical Modeling:*

Mathematical modeling of brain relaxation is performed with least square method. Figs.7 (a)-(c) shows the brain relaxation modeling with mathematical formulation. Equations (6)-(8) shows the brain relaxation modeled expression for Bengali, English and Mathematics test respectively. The normal of the residuals have also been calculated to find the best mathematical model. English has the least norm of the residuals 0.088 indicating that the developed mathematical model is well suited with the original $\beta/\alpha$ Ratio of brain after English test whereas least suited for $\beta/\alpha$ Ratio after Mathematics test. In the case of math reading and math solving the value of the norm of the residual is 0.29392.

$$Y_B = 0.24x_t^2 - 0.9x + 1.3 \quad (6)$$

$$Y_M = 0.27x_t^2 - 1.1x + 1.6 \quad (7)$$

$$Y_E = 0.15x_t^2 - 0.62x + 1.1 \quad (8)$$

TABLE II. Stress Identification for Different Test Environments

| Task | Type | | $\alpha$ | $\beta$ | $\beta/\alpha$ | % of $\alpha$ | % of $\beta$ | Average % of $\alpha$ | Average % of $\beta$ | Mental State |
|---|---|---|---|---|---|---|---|---|---|---|
| Without mental task | Relax | | 4.2 | 2.13 | 0.509 | 66 | 33.7 | 66.26 | 33.73 | Relax |
| With mental task | Bengali | Bengali reading | 1.79 | 2.62 | 1.463 | 40.5 | 59.4 | 41.41 | 58.58 | Stress |
| | | Bengali writing | 2.40 | 3.28 | 1.367 | 42.2 | 57.7 | | | |
| | English | English Reading | 2.45 | 2.94 | 1.198 | 45.4 | 54.5 | 46.46 | 53.53 | Stress |
| | | English Writing | 2.97 | 3.29 | 1.10 | 47.4 | 52.5 | | | |
| | Math | Math reading | 2.07 | 3.685 | 1.77 | 36.0 | 63.9 | 35.67 | 64.32 | Stress |
| | | Math solving | 1.88 | 3.455 | 1.83 | 35.3 | 64.6 | | | |

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on July 20,2025 at 02:40:20 UTC from IEEE Xplore. Restrictions apply.

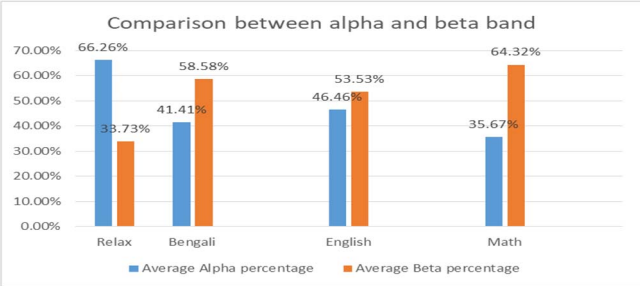



Fig. 5. Percentage Activation of Alpha and Beta band for relax and task oriented stage.

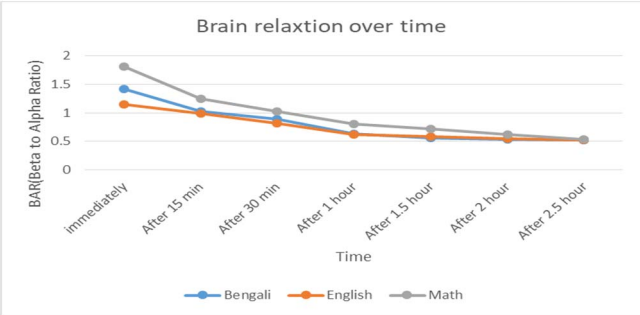
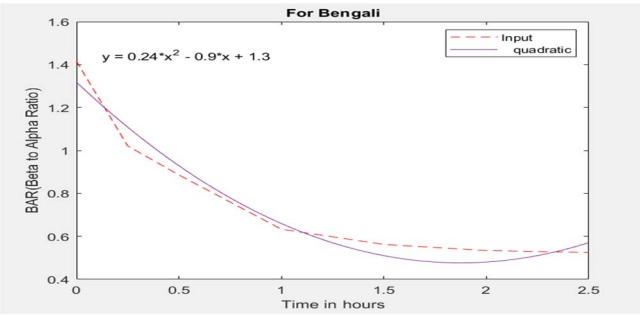
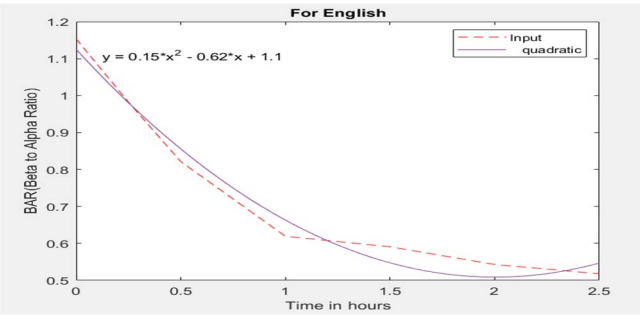


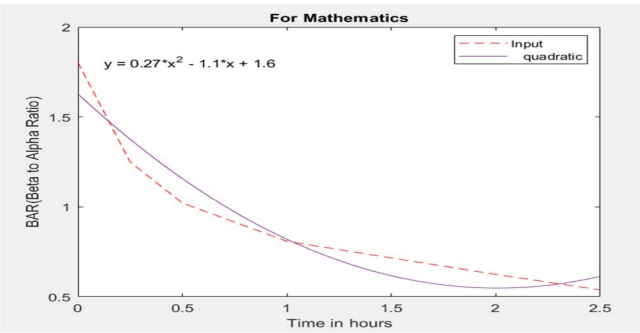

Fig. 6. Brain relaxation over time for various mental tasks.




(a)




(b)




(c)

Fig. 7. Relaxation over time for various mental tasks.

TABLE III. BRAIN RELAXATION ANALYSIS WITH $\beta/\alpha$ RATIO

| Mental Task Types | Just after Task | After 15 min | After 30 min | After 1 hour | After 1.5 hour | After 2 hours | After 2.5 hour |
|---|---|---|---|---|---|---|---|
| Bengali | 1.41 | 1.02 | 0.88 | 0.633 | 0.562 | 0.533 | 0.524 |
| English | 1.15 | 0.98 | 0.821 | 0.619 | 0.590 | 0.542 | 0.517 |
| Math | 1.80 | 1.24 | 1.021 | 0.809 | 0.716 | 0.624 | 0.538 |

TABLE IV. THE COEFFICIENTS OF THE MATHEMATICAL MODEL

| Test | $a_2$ | $a_1$ | $a_0$ | Norm of the residuals |
|---|---|---|---|---|
| Bengali | 0.24 | -0.9 | 1.3 | 0.16515 |
| English | 0.15 | -0.62 | 1.1 | 0.088244 |
| Math | 0.27 | -1.1 | 1.6 | 0.29392 |

## IV. CONCLUSION

Learning of the learners depends on their cognitive capability which affects negatively due to stress. Stress of human brain is the unexpected brain condition which hampers brain response or human behavior. In this paper, stress is identified from the EEG band activation and beta to alpha ratio. The larger PSD value of beta band than alpha indicates the state of stress during different test environment. The beta to alpha band power ratio for relax state of the brain determine that alpha band is more activated for relaxation which maintains an inverse relationship due to stress. After performing the test, the beta and alpha band activation is observed over time from the beta to alpha ratio. Brain relaxation is obtained after 2.5 hours approximately after performing the test. Finally brain relaxation is modeled with mathematical formulation and quadratic mathematical expression is obtained for the brain relaxation modeling after test. The normal of the residuals shows the deviation of our proposed model. So, this work shows an innovative approach of mental stress and brain relaxation modeling for sustained mental task environment.

## ACKNOWLEDGMENT

This research work was partially supported by UGC, Bangladesh. Special thanks are given to all of them who have given their kind consent to take data and use their image for this research activity.

## REFERENCES

[1] JA Fredricks, PC Blumenfeld, and AH Paris, “School engagement: Potential of the concept, state of the evidence. Review of educational research”, vol.74, no.1, 2004, pp.59–109.

[2] CD Wickens, “Multiple resources and mental workload. Human Factors: The Journal of the Human Factors and Ergonomics Society”, vol.50, no.3, 2008, pp. 449–455.

[3] So WKY, SWH Wong, JN Mak, and RHM Chan, (2017) “An evaluation of mental workload with frontal EEG”, PLoS ONE, vol. 12, no.4, 2017, pp. 1-17.

[4] T. H. Holmes and R. H. Rahe, “The social readjustment rating scale”, Journal of Psychosomatic Research, vol. 11, 1967,pp. 213-218.

[5] T. K. Liu, Y. P. Chen, Z.Y. Hou, C. C. Wang, and J.H. Chou, “Noninvasive evaluation of mental stress using by a refined rough set technique based on biomedical signals”, Artificial Intelligence in Medicine, vol. 61, 2014, pp. 97-103.

[6] R. Zheng, S. Yamabe, K. Nakano, and Y. Suda, “Biosignal analysis to assess mental stress in automatic driving of trucks: Palmar perspiration

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on July 20,2025 at 02:40:20 UTC from IEEE Xplore. Restrictions apply.

and masseter electromyography", Sensors, vol. 15, no. 3, 2015, pp. 5136-5150.

[7] A. A. Saidatul, M P. Paulraj, S. Yaacob and Y. Mohd Ali, R. Fadzly, and M. Fauziah, "Spectral Density Analysis: Theta Wave as Mental Stress Indicator", Journal of Communications in Computer and Information Science, vol.260, January 2011, pp.103-112.

[8] J. R. Stroop, "Studies of interference in serial verbal reactions", Journal of Experimental Psychology, vol. 28, 1935, pp.643-662.

[9] K. Dedovic, R. Renwick, N. K. Mahani, V. Engert, S. J. Lupien, and J. C. Pruessner, "The Montreal Imaging StressTask: Using functional imaging to investigate the effects of perceiving and processing psychosocial stress in the human brain", Journal of Psychiatry and Neuroscience, vol. 30, 2005, pp. 319-325.

[10] W. Klimesch, "EEG alpha and theta oscillations reflect cognitive and memory performance: a review andanalysis", Brain research reviews,vol.29, no.2,pp.169–195.

[11] S. Raghavachari, M.J. Kahana, D.S. Rizzuto, J.B. Caplan, M.P. Kirschen, B. Bourgeois, J.R. Madsen, and J.E. Lisman, "Gating of human theta oscillations by a working memory task", Journal of Neuroscience, vol. 21, no.9, 2001, pp.3175-3183.

[12] M. Islam and M. Ahmad, "Modeling of Human Emotion with Effective Frequency Band during a Test of Sustained Mental Task", IEEE International Conference on Electrical and Computer Engineering (WIECON-ECE), pp.403-406, 2015.

Authorized licensed use limited to: UNIVERSIDADE FEDERAL DO RIO GRANDE DO SUL. Downloaded on July 20,2025 at 02:40:20 UTC from IEEE Xplore. Restrictions apply.